

EECS/BioE/MechE 106A/206A

Project Proposal Spring 2026

1 Team Information

Name	Email	Background
Alex Aney	alex_ane@berkeley.edu	Alex is pursuing his undergraduate degree in EECS and Math and loves how robotics allows math and theory to come to life. In his free time he loves building new things.
Shu Yuan	shuoyuan@berkeley.edu	Shu is a senior student in CS and Applied Mathematics. He is actively conducting research in LLM agents and LLM post-training. He is also interested in applying deep learning in robotics.
Natalie Wang	nataliewang@berkeley.edu	Natalie is a second-year studying Mechanical Engineering. She is interested in pursuing a career in surgical robotics in the future.
Yolanda Wei	yiran_wei@berkeley.edu	Yolanda is pursuing her undergraduate degree in ECE and Math. She is interested in applying robotics and machine learning to solving societal issues.

2 Abstract

We propose a robotic system that observes an unfinished math equation on a whiteboard, determines the missing solution, and writes the answer back onto the board. The project combines sensing to perceive the equation and marker, planning to interpret and solve the equation while generating grasping and writing motions, actuation to control the robot arm and gripper, and hardware through a real camera, robot, whiteboard, and marker setup.

3 Project Description

What are your project goals?

Our goals are to build an end-to-end robotic system that can observe an unfinished mathematical equation written on a whiteboard, determine the missing solution, and physically write the answer back onto the board. More specifically, we want the robot to (1) perceive the equation and relevant objects in the scene, (2) reason about the mathematical content of the equation with an external model, (3) locate and grasp a whiteboard marker, and (4) execute a writing motion that places the solution in the correct location on the board. This project is intended to demonstrate an integrated real-world robotics pipeline combining perception, reasoning, motion generation, and manipulation.

How will you design your project in terms of software/hardware architecture (at a high level)?

On the software side, we design four main modules. First, a perception module will process camera images to detect the whiteboard, estimate the writing region, recognize the handwritten equation, and localize the marker. Second, a VLM module will interpret the incomplete equation and generate the missing answer, using an vision-language model. Third, a planning module will convert the output of perception and reasoning into robot actions, including task sequencing, grasp planning, and generation of a writing trajectory in the whiteboard frame. Finally, a control and execution module will command the robot arm and gripper to pick up the marker and write the computed answer onto the board. The characters and the figures from 0 to 9 can be hard-coded as actions into the system.

The hardware setup will consist of a real robot arm, a gripper, a camera, a physical whiteboard, and a whiteboard marker. We currently expect to use a UR7e and a camera such as a Logitech webcam or Realsense depending on calibration and depth requirements. The robot will operate in a fixed workspace where the whiteboard and marker are placed in known approximate regions.

How does your project incorporate sensing, planning, and actuation?

Sensing: The camera will capture an image of the board, and the system will identify the equation, the open space where the solution should be written, and the location and pose of the marker.

Planning: At the task level, the robot must decide the sequence of actions: perceive the board, infer the missing mathematical result, locate the marker, grasp it, move to the board, and write the answer. At the motion level, the robot must plan arm motions for reaching and grasping the marker from the location informed by the sensing module, moving to the writing pose, and tracing the strokes needed to write the answer.

Actuation: The robot arm and gripper will perform the grasping action on the marker, maintain a stable grasp during motion, and execute a controlled writing trajectory against the whiteboard surface.

How will you test or assess your project? What constitutes a success? What are some realistic goals? What are some reach goals?

At the system level, success will mean that the robot can start from an image of an unfinished equation on a real whiteboard and complete the full loop: read the equation, infer the correct answer, pick up the marker, and write the answer back in the correct place with no manual intervention. The system should be consistently working regardless of the equation and the position of the white boards and the marker. Some realistic goals may include picking up the marker and writing on the board with markers and white boards being in a fixed pose; and a natural extension of that (a potential reach goal) might be making the robot to pickup the markers regardless of the pose and the orientation of the marker.

4 Tasks

1. **System design.** We will finalize the overall pipeline, including perception, reasoning, grasping, and writing, and define the interfaces between modules. [by 3/18]
2. **Perception.** We will develop the sensing pipeline for the whiteboard scene.
 - (a) **Calibration.** Calibrate the camera, robot, and whiteboard frames. [by 3/31]
 - (b) **Equation recognition.** Detect and read the unfinished equation from the whiteboard image. [by 3/31]
 - (c) **Marker detection.** Detect and localize the marker for grasping. [by 3/31]
3. **Reasoning and planning.** We will convert perception outputs into robot actions.
 - (a) **Equation solving.** Use an LLM to infer the missing solution. [by 3/31]
 - (b) **Task planning.** Implement the sequence: read, solve, find marker, grasp, and write. [by 4/7]
 - (c) **Writing trajectory generation.** Convert the answer into robot writing motions. [by 4/14]
4. **Manipulation and control.** We will enable the robot to pick up the marker and write.
 - (a) **Marker grasping.** Develop and test reliable grasping motions. [by 4/21]
 - (b) **Writing execution.** Execute writing motions on the board. [by 4/21]
5. **Checkpoint 1.** Demonstrate whiteboard image capture, equation recognition, equation solving, and marker localization. [by 4/21]
6. **Integration and testing.** We will combine all modules and evaluate the full system.
 - (a) **System integration.** Integrate perception, planning, and control into one pipeline. [by 4/28]

- (b) **Checkpoint 2.** Demonstrate an end-to-end system that reads an equation, picks up the marker, and writes the answer. [by 4/28]

7. **Presentation and Demos.** [by 5/8]

8. **Report.** [by 5/15]

5 Bill of Materials

5.1 Use of Lab Resources

Please include all lab resources you plan to use so we can ensure that all teams have sufficient access to hardware. Please indicate which robot end effectors / grippers you plan to use, if applicable.

Item	Quantity
UR7e (w/ parallel gripper)	1
RealSense Cameras	1

5.2 Other Purchases

Remember to fill out the form on the Final Project Guidelines if you want ESG to purchase anything for you with your \$55 budget! Anything beyond that will be up to you to buy.

Item	Quantity	Justification
Whiteboard	1	Necessary for the project
Markers	5	Necessary for the project